A Digital-Domain Calibration of Split-Capacitor DAC for a Differential SAR ADC Without Additional Analog Circuits

Abstract—**A digital-domain calibration method is proposed for a split-capacitor DAC (split-CDAC) used in a differential-type 11-bit SAR ADC. It calibrates the nonlinearities of SAR ADC due to the DAC capacitance mismatch as well as the two parasitic capacitances connected in parallel with each of the bridge capacitor and the LSB bank of split-CDAC. The proposed ADC does not require any additional analog circuits for calibration, because it utilizes one of the two split-CDACs to measure the error codes of the other split-CDAC. During the normal A/D conversion step, the 11.5-bit raw SAR code output of ADC is added to the pre-measured error codes to generate the 11-bit calibrated output code. The analog block of the ADC was fabricated in a 0.13-μm CMOS process, and the digital block was implemented in a FPGA. The measured SNDR and SFDR are 61.6 dB (ENOB 9.93 bits) and 78 dB at the Nyquist rate with a 5 kHz sine wave input. INL and DNL are measured to be $+0.96/-0.98$ LSB, and $+0.96/-0.97$ LSB, respectively. This work extends the prior work by utilizing an additional 0.5-bit raw SAR code to eliminate the missing code, and by employing a temporal averaging with a FIR LPF to measure the error code reliably in spite of the supply noise.**

Index Terms—**Analog-to-digital converter (ADC), digital-domain calibration, split-capacitor digital-to-analog converter (DAC), successive approximation register (SAR).**

I. Introduction

THE low-voltage and low-power sensor interface circuits are required for the portable biopotential monitoring devices [1]–[4]. In general, a biopotential signal such as electrocardiogram (ECG) has the amplitude between tens of μV and several mV, which corresponds to the dynamic range of 50–60 dB. A SAR ADC with the ENOB of 9–10 bits is suitable for the above mentioned devices.

Usually, the DAC block of SAR ADC is implemented by using a binary weighted capacitor array, which occupies most of chip area of SAR ADC. To reduce the DAC chip area, a split-CDAC may be used [5]–[12]. In the split-CDAC, a binary weighted capacitor array is divided into two smaller binary weighted capacitor arrays, which are connected by using a bridge capacitor [5]–[12]. For example, a 12-bit binary weighted capacitor array can be divided into two 6-bit binary weighted capacitor arrays, where the DAC chip area is reduced to 1/32 of the original DAC chip area. However, the split-CDAC is vulnerable to the nonlinearity distortion, which is caused by the parasitic capacitances associated with the bridge capacitor as well as the mismatches of the binary weighted capacitor arrays [7], [8]. The two capacitor arrays in a split-CDAC are divided into the MSB array and the LSB array. The MSB array is connected to the input node of comparator. The LSB array is connected to the MSB array through a bridge capacitor. Two kinds of parasitic capacitors degrade the linearity of split-CDAC. One is the parasitic capacitance of LSB array. The other is the parasitic capacitance in parallel with the bridge capacitor [7], [8].

Recently, some calibration methods to enhance linearity of split-CDAC have been proposed [7], [8]. In [7], a tunable capacitor in parallel with the LSB array is adjusted to make the LSB capacitance of MSB array to be equal to the capacitance seen from the MSB array toward the LSB array through the bridge capacitor. Although this calibration method compensates for the nonlinearities due to parasitic capacitors, it cannot compensate for the nonlinearity due to the capacitance mismatch among DAC capacitors.

In [8], the nonlinearities due to both the parasitic capacitors and the capacitance mismatch among DAC capacitors are compensated. A tunable capacitor compensates for the nonlinearity due to parasitic capacitors, as in [7]. An extra calibration DAC is used to compensate for the nonlinearity due to the capacitance mismatch among DAC capacitors. The added calibration DAC of [8] measures the DAC capacitance mismatch of the three most significant bits of split-CDAC by utilizing the self-calibration technique [13]. During the normal A/D conversion step, the calibration DAC performs the nonlinearity compensation in the analog domain. Since the calibration DAC is implemented in analog circuits, the linearity and the monotonic characteristic of the calibration DAC is crucial to achieve the good performance of the entire ADC. As a result, a careful layout technique is required for the calibration DAC.

In this work, a digital-domain calibration method is proposed to compensate for the nonlinearities due to both the DAC capacitance mismatch and the parasitic capacitances of the LSB array and the bridge capacitor. No additional calibration DAC is required in this work. Since the differential architecture is used for

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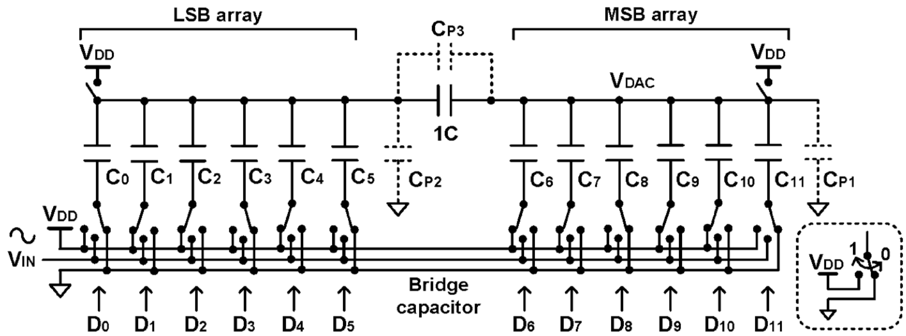

Fig. 1. 11.5-bit split-CDAC with parasitic capacitors.

the SAR ADC of this work, there are two parallel branches of split-CDACs. The nonlinearity of one branch of split-CDAC is measured by the other branch of split-CDAC, and the results are stored as digital error codes. During the normal A/D conversion step, the raw output code of SAR ADC is added to the pre-measured digital error codes to generate the nonlinearity-compensated output code. In the digital calibration scheme of ADC [14], [15], no additional analog block is used for calibration. Only digital block is added for calibration.

This paper is organized as follows. Section II describes the analysis of nonlinearity of split-CDAC and the calibration principle. Section III presents the architecture of the implemented ADC, the error-code measurement method of split-CDAC and the offset calibration scheme of comparator. Measurement results and conclusion follow in Section IV and V, respectively.

II. A Digital Calibration Method for Split-CDAC

A. Nonlinearity of Split-CDAC

Fig. 1 shows the 11.5-bit split-CDAC with the parasitic capacitors (C_{P1}, C_{P2}, C_{P3}). The parasitic capacitors, such as C_{P1} and C_{P2}, are caused by overlaps between the routing metal layout and the silicon substrate. The parasitic capacitor C_{P3} is caused by the overlaps among the metal layouts. The parasitic capacitor C_{P1} does not affect the charge redistribution of split-CDAC, because the charge of C_{P1} at the end of the A/D conversion step is the same as that at the start time of conversion step. That is, the output node V_{DAC} of split-CDAC is connected to the supply voltage V_{DD} during the initial sample phase of the A/D conversion step and converges to the supply voltage V_{DD} at the end of A/D conversion step. On the other hand, the charge amounts of the parasitic capacitors C_{P2} and C_{P3} are not the same at the start time and the end time of the A/D conversion step. Moreover, the charge difference of each parasitic capacitor C_{P2} and C_{P3} between the beginning and end-time points of A/D conversion step depends on the input analog voltage V_{IN}. This gives the signal (V_{IN}) dependent distortion, which is nonlinearity [7], [8].

All the capacitors $(C_0, C_1, \cdots, C_{11})$ of the LSB and MSB arrays in the split-CDAC of Fig. 1 are implemented by the parallel connection of unit capacitors. Let C be the average value of all the unit capacitors which comprise the DAC capacitors $(C_0, C_1, \cdots, C_{11})$.

$$C = \frac{\sum_{i=0}^{11} C_i}{2 \cdot (2^6 - 1)} \tag{1}$$

The parasitic capacitors (C_{P1}, C_{P2}, C_{P3}) are not included in the C calculation of (1). As a result, the capacitance of each DAC capacitor can be expressed as

$$C_i = \begin{cases} 2^i C \cdot (1+\varepsilon_i), & i = 0, \cdots, 5 \\ 2^{i-6} C \cdot (1+\varepsilon_i), & i = 6, \cdots, 11 \end{cases} \tag{2}$$

In (2), $\varepsilon_\imath$ $(i = 11, \cdots, 0)$ represents the mismatch ratio from the desired value of each DAC capacitor C_i. The nominal capacitance of the bridge capacitor is set to the average capacitance C of the unit capacitors. The difference in capacitance of the bridge capacitor from the average capacitance C is taken as C_{P3}.

During the sample phase of A/D conversion step, the top plates of all the DAC capacitors $(C_{11}, C_{10}, \cdots, C_0)$ are connected to V_{DD} and the bottom plates are connected to V_{IN}. Thus, all the DAC capacitors sample the voltage of $(V_{DD} - V_{IN})$. During the conversion phase, when the top plates of all the DAC capacitors are floated and the bottom plates are connected to V_{DD} or GND depending on the code values $(D_{11}, D_{10}, \cdots, D_0)$, the output voltage V_{DAC} of split-CDAC can be expressed as

$$\begin{aligned} V_{DAC} = V_{DD} - V_{IN} + \frac{V_{DD}}{2^{11}}(2^{10} \cdot D_{11} + \cdots + D_1 + 0.5D_0) \\ +(E_{V.11} \cdot D_{11} + \cdots + E_{V.1} \cdot D_1 + E_{V.0} \cdot D_0) \end{aligned} \tag{3}$$

where the supply voltage V_{DD} is used as the reference voltage of split-CDAC. In (3), D_i $(i = 11, \cdots, 0)$ is the input digital logic value of the corresponding DAC capacitor $(C_{11}, \cdots, C_0)$ in Fig. 1. D_i is replaced by an integer value of either 1 or 0 in (3) when the digital logic value is high or low, respectively. $E_{V.i}$ $(i = 11, \cdots, 0)$ represents the error voltage of each DAC capacitor $(C_{11}, \cdots, C_0)$ due to the parasitic capacitance (C_{P2}, C_{P3}) and the mismatch of the i-th DAC capacitance, $\varepsilon_\imath$ $(i = 11, \cdots, 0)$. When the logic value of D_i $(i = 11, \cdots, 0)$ is set to "1," the corresponding DAC capacitor C_i $(i = 11, \cdots, 0)$ changes the output voltage V_{DAC} of split-CDAC by $V_{DD}/2^{12-i} + E_{V.i}$. When D_i is "0," C_i does not change V_{DAC}. The equation for $E_{V.i}$ can be derived as (4a) by translating the LSB array capacitors to the MSB array side [7] and calculating the contribution of each capacitor (C_i) to the split-CDAC output (V_{DAC}). (4a) can be simplified to (4b) by assuming $2^6 \cdot C \gg (C_{P2} + 2 \cdot C_{P3})$. (See (4a)–(4b) at the bottom of the next page.)

Even in the ideal case of no mismatch $(\varepsilon_\imath = 0)$ and no parasitic capacitors $(C_{P2} = C_{P3} = 0)$, the error voltage $E_{V.i}$ has non-zero value of $2^{i-24} \cdot V_{DD}$. This non-zero error voltage of ideal case is generated by the systematic offset (gain error) of the split-CDAC architecture shown in Fig. 1. The equivalent capacitance of the LSB array in series with the bridge capacitor is not exactly C but $C\cdot(1-2^{-6})$, which generates the above-mentioned systematic offset [7]. The sum of all error voltage $E_{V.i}$ $(i = 11,\cdots,0)$ is 0.5 LSB, because the sum of $2^{i-24}\cdot V_{DD}$ from $i=0$ to 11 is around $2^{-12}\cdot V_{DD}(=0.5\text{ LSB})$.

The dynamic performance (ENOB) of the SAR ADC with split-CDAC is estimated through Monte Carlo simulation (MATLAB behavioral simulation), where the parasitic capacitance (C_{P2}, C_{P3}) and the DAC capacitance mismatch ε_i are considered. However, the input-referred noise voltage of comparator and the kT/C noise of DAC are not included in the Monte Carlo simulation. In the Monte Carlo simulation, the capacitance value of each DAC capacitor C_i $(i = 11,\cdots,0)$ is determined by assuming that the DAC capacitance mismatch ε_i has a Gaussian distribution $(3\sigma = 0.1\%)$. In order to achieve an ENOB of 10 bits with the probability of 99.9% for a given DAC capacitance mismatch $(3\sigma = 0.1\%)$, the required minimal value of parasitic capacitance of C_{P2} and C_{P3} are 3 C and 0.085 C, respectively. When the input-referred noise voltage of comparator is 0.25 LSB, an ENOB is reduced to around 9.5 bits [9], [16]. Therefore it is required to calibrate the split-CDAC for an ENOB larger than 10 bits.

B. Digital-Domain Calibration

The SAR ADC of this work has a differential architecture to accept the differential analog voltage as input for A/D conversion. Thus, the DAC block consists of two branches of split-CDAC. One is the positive-branch split-CDAC and the other is the negative-branch split-CDAC. When one branch split-CDAC works independently of the other branch [11], the output voltage V_{DACP} of the positive branch split-CDAC can be expressed as

$$V_{DACP} = V_{DD}-V_{INP}+\frac{V_{DD}}{2^{11}}(2^{10}\cdot D_{11}+\cdots+D_1+0.5\cdot D_0) +(E_{VP.11}\cdot D_{11}+\cdots+E_{VP.1}\cdot D_1+E_{VP.0}\cdot D_0) \quad (5)$$

where $E_{VP.i}(i = 11,\cdots,0)$ represents the error voltage of each DAC capacitor $(C_{11},\cdots,C_0)$ of the positive-branch split-CDAC, as shown in (4). To calculate the output voltage V_{DACN} of the negative-branch split-CDAC, $1-D_i$ and $E_{VN.i}$ are substituted into D_i and $E_{VP.i}$ of (5), respectively. The fact that the sum of all error voltage $(\sum_{i=0}^{11} E_{VN.i})$ is 0.5 LSB is also used in the derivation step.

$$V_{DACN} = 2V_{DD}-V_{INN}-\frac{V_{DD}}{2^{11}}(2^{10}\cdot D_{11}+\cdots+D_1+0.5\cdot D_0) -(E_{VN.11}\cdot D_{11}+\cdots+E_{VN.1}\cdot D_1+E_{VN.0}\cdot D_0) \quad (6)$$

Since the output voltage of two split-CDACs converges to the same value at the end of A/D conversion step, the differential analog input voltage $(V_{INP}-V_{INN})$ can be related to the digital output code $(D_{11}, D_{10},\cdots,D_0)$ as

$$\left(\frac{V_{INP}-V_{INN}}{2}\right)=\frac{V_{DD}}{2^{11}}(2^{10}\cdot D_{11}+\cdots+D_1+0.5\cdot D_0)-\frac{V_{DD}}{2}+\frac{1}{2} \cdot[(E_{VP.11}+E_{VN.11})\cdot D_{11}+\cdots+(E_{VP.0}+E_{VN.0})\cdot D_0] \quad (7)$$

Out of the 11.5-bit digital code $(D_{11}, D_{10},\cdots,D_1,D_0)$, only 11 bits $(D_{11}, D_{10},\cdots,D_1)$ are used as the final ADC output code. The D_0 bit is added in the ADC output code to represent a fraction (0.5 LSB), which is used for the error correction purpose only. From (7), the raw SAR code, which is the ADC output code with no calibration, can be written as

$$\begin{aligned} Raw\ SAR\ Code &= 2^{10}+\frac{V_{INP}-V_{INN}}{2}\cdot\frac{2^{11}}{V_{DD}} \\ &\quad -(E_{C.11}\cdot D_{11}+\cdots+E_{C.0}\cdot D_0) \\ &= 2^{10}\cdot D_{11}+2^9\cdot D_{10}+\cdots+D_1+0.5 \\ &\quad \cdot D_0 \end{aligned} \quad (8a)$$

$$E_{C.i} = \frac{1}{2}\cdot(E_{VP.i}+E_{VN.i})\cdot\frac{2^{11}}{V_{DD}} \quad (8b)$$

where $E_{C.i}$ $(i = 11,\cdots,0)$ is the error code, which is the digital value for the average error voltage $(0.5\cdot(E_{VP.i}+E_{VN.i}))$ of C_i in the positive and negative branches of split-CDAC. In (8a), the raw SAR code has some discrepancy from the ideal value $\lfloor 2^{10}+0.5\cdot(V_{INP}-V_{INN})\cdot 2^{11}/V_{DD}\rfloor$. The discrepancy corresponds to the error component $(E_{C.11}\cdot D_{11}+\cdots+E_{C.0}\cdot D_0)$.

The corrected code D_{OUT} can be obtained by adding the sum of error codes to the raw SAR code.

$$\begin{aligned} D_{OUT} &= Raw\ SAR\ Code \\ &\quad + Sum\ of\ Error\ Codes \end{aligned} \quad (9a)$$

$$\begin{aligned} Sum\ of\ Error\ Codes &= E_{C.11}\cdot D_{11} \\ &\quad +\cdots+E_{C.0}\cdot D_0 \end{aligned} \quad (9b)$$

Thus, D_{OUT} is equal to the ideal value of $[2^{10}+0.5\cdot(V_{INP}-V_{INN})\cdot 2^{11}/V_{DD}]$, which is independent of the DAC capacitance mismatch ε_i and the parasitic capacitances (C_{P2}, C_{P3}).

The procedure of the error correction is as follows. Before the normal A/D conversion step starts, the analog error voltage $(E_{VP.i}, E_{VN.i}, i = 11,\cdots,6)$ of each bit D_i is converted to the digital error code $(E_{C.i}, i = 11,\cdots,6)$. The digital error code is stored in a memory. It consists of 4 regular bits and 2 fractional bits (0.5, 0.25 LSB), because the output of the lower 5.5-bit ADC used for the error voltage measurement is divided by 2

$$\frac{V_{DD}}{2^{12-i}}+E_{V.i} \cong \begin{cases} \frac{2^i\cdot(1+\varepsilon_i)\cdot C+2^i\cdot C_{P3}}{(2^{12}-1)\cdot C+(2^6-1)\cdot(C_{P2}+2\cdot C_{P3})}V_{DD}, & i=0,\cdots,5 \\ \frac{2^i(1+\varepsilon_i)\cdot C+2^{i-6}\cdot(C_{P2}+C_{P3})}{(2^{12}-1)\cdot C+(2^6-1)\cdot(C_{P2}+2\cdot C_{P3})}V_{DD}, & i=6,\cdots,11 \end{cases} \quad (4a)$$

$$E_{V.i} \cong \begin{cases} \frac{2^{i-12}+2^i\cdot\varepsilon_i+\frac{2^i\cdot(C_{P3}-2^{-6}\cdot C_{P2})}{C}}{2^{12}}\cdot V_{DD}, & i=0,\cdots,5 \\ \frac{2^{i-12}+2^i\cdot\varepsilon_i-\frac{2^{i-6}\cdot(2^{-12}\cdot C_{P2}+C_{P3})}{C}}{2^{12}}\cdot V_{DD}, & i=6,\cdots,11 \end{cases} \quad (4b)$$

TABLE I
ERROR VOLTAGE IN LSB FROM (4b) ($C_{P2} = 500$ fF, $C_{P3} = 6$ fF, $C = 100$ fF, $\varepsilon = 1/2^8$)

i	0	1	2	3	4	5
$\lvert E_{V.i} \rvert_{\max}$	0.01	0.01	0.03	0.06	0.11	0.22
i	6	7	8	9	10	11
$\lvert E_{V.i} \rvert_{\max}$	0.10	0.20	0.41	0.82	1.64	3.27

during the error code calculation. When an ADC begins to perform the normal A/D conversion, each bit D_i $(i = 11, \cdots, 0)$ of the 11.5-bit raw SAR code is sequentially determined from a bit D_{11} (MSB) to a bit D_0. In the process of the normal A/D conversion, if a bit D_i $(i = 11, \cdots, 6)$ of the six MSBs of raw SAR code is determined to the logic value "1," the corresponding error code $E_{C.i}$ is fetched from a memory and summed with the 11.5-bit raw SAR code. The error correction is completed with the decision of a bit D_0.

Although all bits of the 11.5-bit raw SAR code are used for calibration in (9), only six MSB codes $(D_{11}, D_{10}, \cdots, D_6)$ are used for calibration in the real implementation. The six LSB codes $(D_5, D_4, \cdots, D_0)$ are not needed for calibration because the sum of error codes for the six LSB codes is smaller than 0.5 LSB. This is guaranteed because the DAC capacitors satisfy the 8-bit resolution and the effect of the parasitic capacitors (C_{P2}, C_{P3}) is smaller than the 10-bit resolution. This is verified in Table I, where the error voltage $E_{V.i}$ for the i-th bit is calculated in LSB units. The worst case sum of error codes for the six LSB codes is 0.44 LSB, which is smaller than 0.5 LSB.

In the error correction, both of [11] and this work use the error codes of six MSBs $(D_{11}, \cdots, D_6)$ of the raw SAR code. The difference between [11] and this work is the number of bits of raw SAR code, which is utilized in the error correction. That is, the prior work [11] utilizes 11 bits of raw SAR code, and this work utilizes 11.5 bits of raw SAR code in the error correction.

The 11-bit ADC has missing codes if the number of output code bin is smaller than 2^{11} for a full-scale analog input. The prior work [11] cannot remove missing codes, because it cannot increase the number of output code bin by error correction. Since the error code $E_{C.i}$ in Fig. 2 is a fixed constant, the corrected code D_{OUT} is deterministically determined by the raw SAR code. That is, the raw SAR code directly determines D_{OUT}. Thus, the number of corrected code bin is equal to or smaller than the number of the raw SAR code bin. In this work, an additional bit D_0 corresponds to a 0.5 LSB of the raw SAR code. The raw SAR code with D_0 is added to the selected error codes to get D_{OUT}, as shown in Fig. 2(b). Since D_0 can take one of two values ("0" or "1"), the number of corrected code bin is larger than that of the raw SAR code bin. This reduces the missing code. The addition of D_0 and the selected error codes to the 11-bit raw SAR code eliminates the missing codes almost completely.

For example, assume that there is a missing code in 96 of the 11-bit raw SAR code (D_{RAW}) of the prior work [11]. The missing code is located at the boundary between the code segments, as shown in Fig. 3(a). The code segment represents a set of 32 codes with the same 6 MSB value. The missing code 96 in the 11-bit raw SAR code corresponds to two missing codes in 96 and 96.5 in the 11.5-bit raw SAR code [Fig. 3(b)]. Since the fractional bits are abandoned in the error correction of the prior work, the code 96 remains to be the missing code in the corrected output (D_{OUT}). However, in this work with the 11.5-bit raw SAR code, there is no missing code in D_{OUT} because the code 95.5 in D_{RAW} is corrected to the code 96 in D_{OUT}.

The proposed digital-domain DAC calibration method can be applied to any capacitor DAC with the differential architecture as well as the differential split-CDAC.

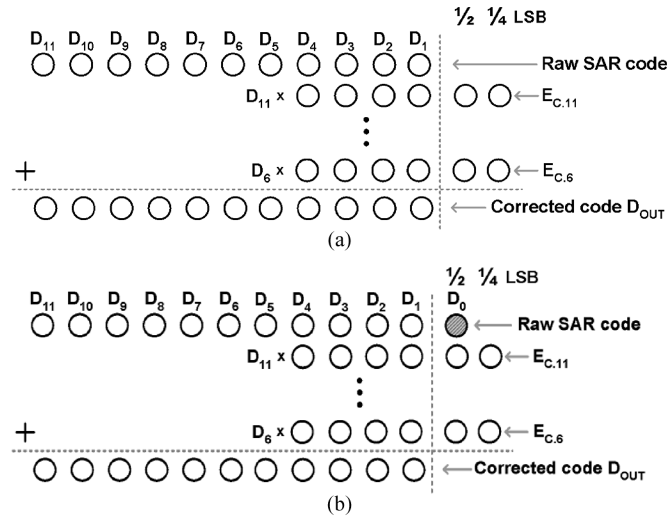

Fig. 2. Error correction methods: (a) prior work [11]; (b) this work.

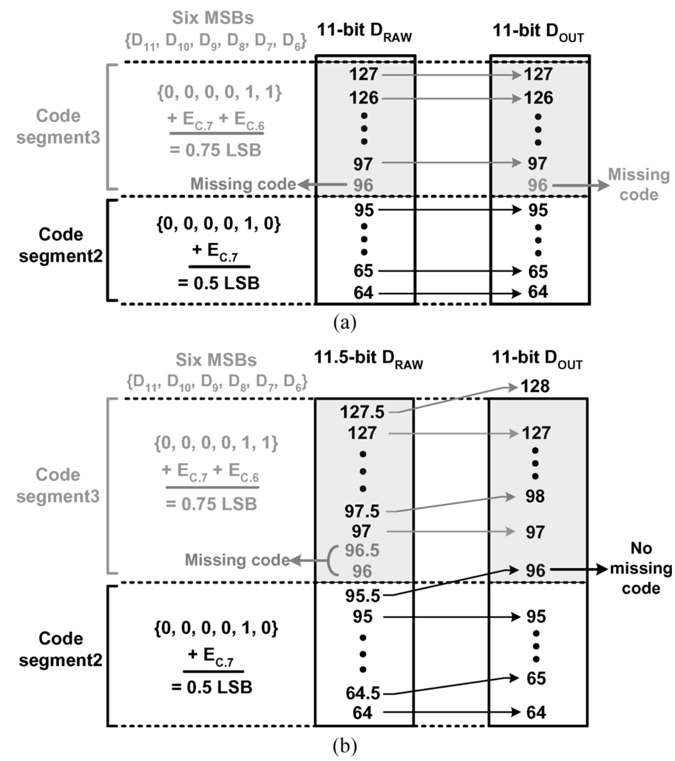

Fig. 3. Example of error correction: (a) prior work; (b) this work.

III. CIRCUIT IMPLEMENTATION

A. Architecture

Fig. 4 shows the architecture of the proposed split-CDAC SAR ADC, which convert the differential input analog voltage

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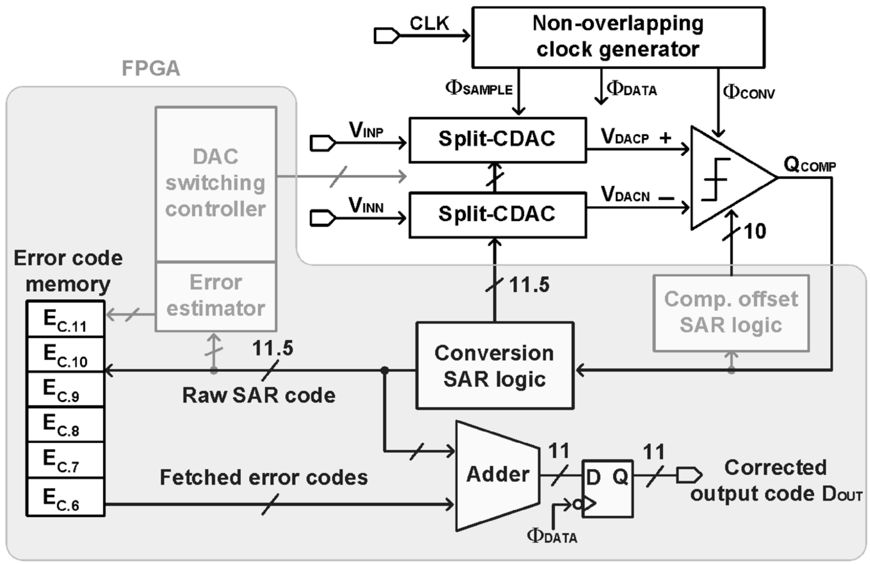


Fig. 4. Architecture of proposed split-CDAC SAR ADC.

$(V_{INP} - V_{INN})$ to the digital code D_{OUT}, by using two split-CDACs, a comparator, a non-overlapping clock generator, a conversion SAR logic, an error code memory and a digital adder. Building blocks such as a DAC switching controller, an error estimator and an error code memory are utilized to estimate the error code of each DAC capacitor. The comparator-offset SAR logic block is used to compensate for the input offset voltage of comparator. The analog building blocks such as two split-CDACs, a comparator and a non-overlapping clock generator were fabricated in a chip. The other digital building blocks were implemented in a FPGA (Vertex5, Xilinx) for the test flexibility.

The operation sequence of ADC is as follows. The first step is the calibration of the input offset voltage of comparator, which is done by using the comparator-offset SAR logic block. As the second step, the error codes $(E_{C.i}, i = 11, \cdots, 6)$ of DAC capacitors are estimated, and the estimated error codes are stored in a memory. After that, the normal A/D conversion step starts by sampling the input analog voltage V_{INP} and V_{INN} on a corresponding split-CDAC, which is driven by the conversion SAR logic block. The output voltage (V_{DACP}, V_{DACN}) from two split-CDACs are compared by a comparator, which is a time-domain comparator [6], [9]. The non-overlapping clock generator generates the sample clock Φ_{SAMPLE}, the conversion clock Φ_{CONV} and the output-data (D_{OUT}) sample clock Φ_{DATA} from an external clock signal CLK with a frequency of 140 kHz. The sampling operation is performed during one period of CLK, while Φ_{SAMPLE} is "1." The SAR operation is performed during the next twelve periods of CLK, while Φ_{CONV} is switching with the same frequency as CLK. Φ_{DATA} remains at "1" only during the next one period of CLK. At the falling edge of Φ_{DATA}, the adder output is sampled as the final data (D_{OUT}) of SAR ADC. Thus, one A/D conversion cycle takes fourteen periods of CLK. The 13-bit adder in FPGA adds the 11.5-bit raw SAR code and the error codes $E_{C.i}$ fetched from an internal memory. The six MSBs out of the 11.5-bit raw SAR code are used to decide whether to add the corresponding error code to the 11.5-bit raw SAR code or not. If D_i $(i = 11, 10, \cdots, 6)$ is "1," the error code $E_{C.i}$ is added to the raw SAR code to generate the corrected output code D_{OUT}.

B. Error Voltage Measurement of Split-CDAC

To find the error code $E_{C.i}$ of (9b) used for the digital-domain DAC calibration, the error voltage due to each DAC capacitor $[E_{VP.i}, E_{VN.i}, (4)]$ is measured for both positive and negative branches of split-CDAC. As shown in (9), the error code $E_{C.i}$ $(i = 11, 10, \cdots, 6)$ should be calculated and stored in a memory prior to the normal A/D conversion step. $E_{C.i}$ is calculated from the measured error voltage $(E_{VP.i}, E_{VN.i})$. $E_{VP.i}$ and $E_{VN.i}$ are the error voltage associated with the i-th DAC capacitor, as shown in (4). $E_{VP.i}$ and $E_{VN.i}$ are converted into digital codes by using the lower 5.5-bit DAC of the other branch split-CDAC, respectively. The average value of the $E_{VP.i}$ and $E_{VN.i}$ codes is calculated and stored as $E_{C.i,}$ as shown in (8b).

Fig. 5 shows the procedure, which is the self-calibration switching scheme [13], to measure $E_{VP.11}$, which is the error voltage associated with the MSB capacitor C_{11} of the positive branch split-CDAC. C'_{11} represents the equivalent capacitance of all the other capacitors of the positive branch split-CDAC except C_{11}, as shown in Fig. 1.

$$C'_{11} = \sum_{i=6}^{10} C_i + \frac{1}{\frac{1}{C+C_{P3}} + \frac{1}{C_{P2}+\sum_{j=0}^{5} C_j}} \tag{10a}$$

$$C'_{11} \cong \sum_{i=6}^{10} C_i + 2^{-6} \cdot \sum_{j=0}^{5} C_j + C_{P3} + \frac{C_{P2}}{2^{12}} \tag{10b}$$

In the ideal case of perfect matching, C'_{11} is nearly the same as C_{11} $(C_{11} - C'_{11} = C/2^6)$. In the real case, a rather large mismatch occurs between C_{11} and C'_{11} because of the DAC capacitance mismatch ε_i and the parasitic capacitance (C_{P2}, C_{P3}).

During the pre-charge phase [Fig. 5(a)], C_{11} is charged to V_{DD} and C'_{11} is discharged to 0, by connecting the top plates of both C_{11} and C'_{11} to V_{DD} and the bottom plates to 0 (GND)

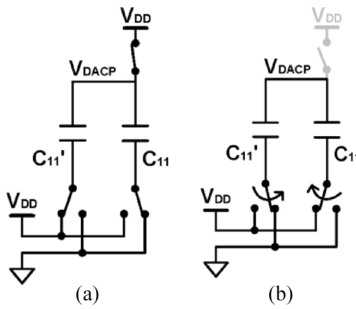


Fig. 5. Error voltage measurement for $E_{VP.11}$: (a) pre-charge phase; (b) evaluation phase.

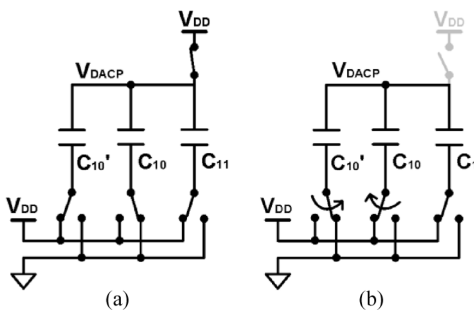


Fig. 6. Error voltage measurement for $E_{VP.10}$: (a) pre-charge phase; (b) evaluation phase.

and V_{DD}, respectively. During the evaluation phase [Fig. 5(b)], the top plates are left floated, and the bottom plates of C_{11} and C'_{11} are connected to V_{DD} and 0, respectively. In the ideal case of $C_{11} = C'_{11}$, the residual voltage (V_{DACP}) of the evaluation phase is very close to V_{DD}. In the real case, $V_{DACP} - V_{DD}$ is non-zero and proportional to $C_{11} - C'_{11}$. The measured $V_{DACP} - V_{DD}$ value during the evaluation phase corresponds to $2E_{VP.11}$, where the term 2 is originated because the switching of the bottom plate of C_{11} from GND to V_{DD} contributes $E_{VP.11}$ to the residual voltage $V_{DACP} - V_{DD}$ and the switching of the bottom plate of C'_{11} from V_{DD} to GND contributes another $E_{VP.11}$ to the residual voltage.

$$V_{DACP} - V_{DD} = 2E_{VP.11} = \frac{C_{11} - C'_{11}}{C_{11} + C'_{11}} \cdot V_{DD} \tag{11}$$

The substitution of (2) for C_{11} and the approximated (10b) for C'_{11} into (11) shows

$$E_{VP.11} \cong 2^{-13} \cdot V_{DD} + 0.5 \cdot \varepsilon_{11} \cdot V_{DD} - 2^{-7} \cdot \frac{(2^{-12} \cdot C_{P2} + C_{P3})}{C} \cdot V_{DD}. \tag{12}$$

This is approximately equal to (4b).

The error voltage ($E_{VP.10}$) due to C_{10} can be measured similarly to $E_{VP.11}$, as shown in Fig. 6. The bottom plate of C_{11} is connected to V_{DD} during both pre-charge and evaluation phase. The bottom plates of C_{10} and C'_{10} are switched in the same way as those of C_{11} and C'_{11} during the $E_{VP.11}$ measurement in Fig. 5. The measured $V_{DACP} - V_{DD}$ during the evaluation phase corresponds to $2E_{VP.10} + E_{VP.11}$, which is proportional to $C_{10} - C'_{10}$ as follows.

$$V_{DACP} - V_{DD} = 2E_{VP.10} + E_{VP.11} = \frac{C_{10} - C'_{10}}{C_{11} + C'_{11}} \cdot V_{DD} \tag{13}$$

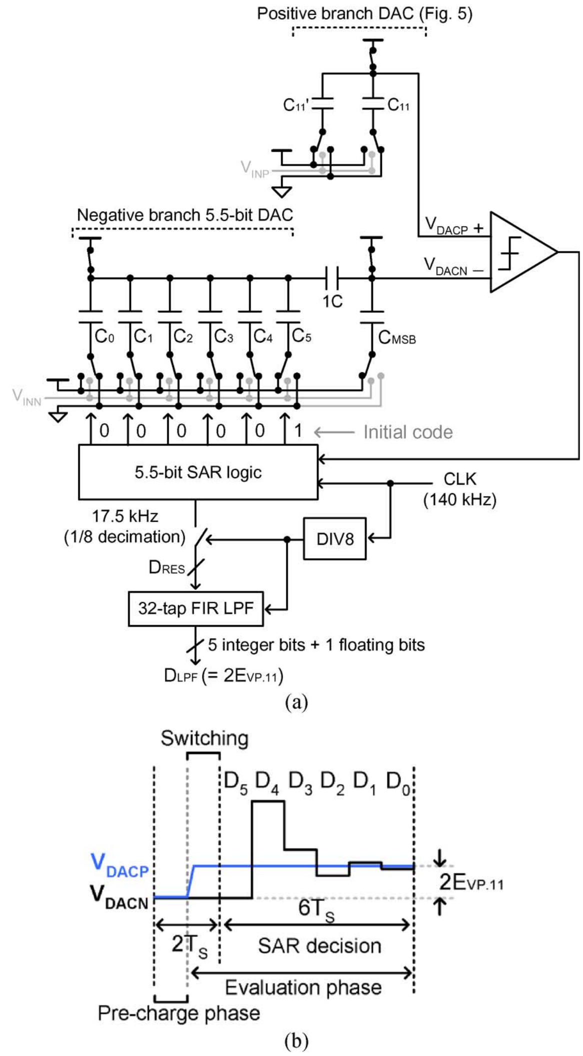


Fig. 7. A/D conversion of error voltage for $E_{VP.11}$: (a) pre-charge phase; (b) waveform of DAC output voltage.

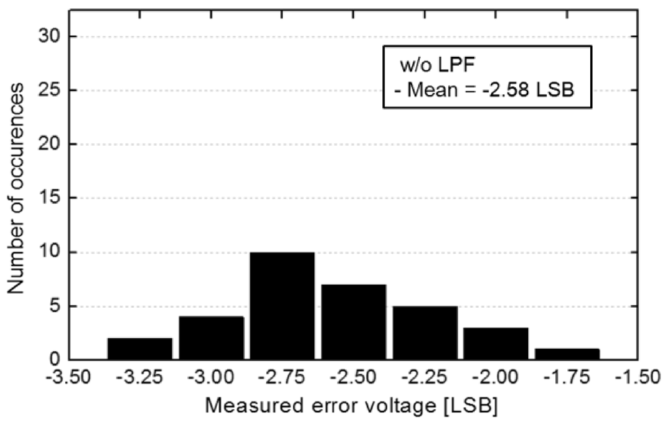
where one $E_{VP.10}$ term in the left side of (13) is contributed by the bottom plate switching of C_{10}, another $E_{VP.10}$ term is contributed by C'_{10}, and the $E_{VP.11}$ term is contributed by $C'_{11}(= C_{10} + C'_{10})$. Similarly the error voltage ($E_{VP.i}$) due to the DAC capacitor C_i ($i = 9, \cdots, 6$) can be derived as

$$V_{DACP} - V_{DD} = 2E_{VP.i} + \sum_{n=i+1}^{11} E_{VP.n} = \frac{C_i - C'_i}{C_{11} + C'_{11}} \cdot V_{DD} \tag{14}$$

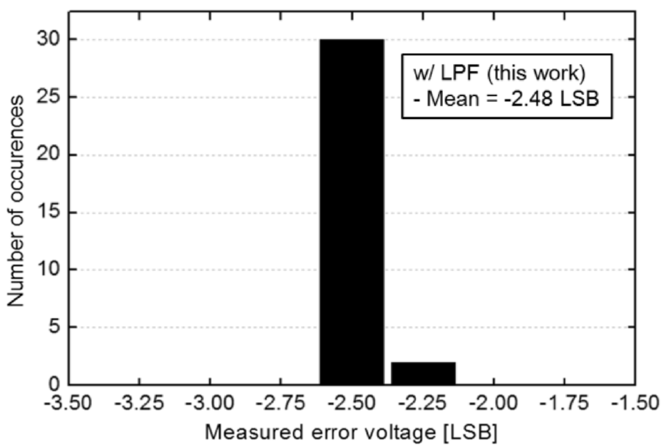
The error voltage $E_{VN.i}$ for the negative branch split-CDAC can be measured similarly to $E_{VP.i}$.

To get the digital value of the measured analog error voltage $E_{VP.11}$, the $V_{DACP} - V_{DD}$ value developed at the evaluation phase of C_{11} switching of Fig. 5 is converted into digital code by the negative branch split-CDAC, as shown in Fig. 7(a). Only the lower 5.5-bit DAC (LSB array) of the negative-branch split-CDAC is used for this conversion. The bottom plates of the upper 6-bit capacitors ($C_6, \cdots, C_{11}$) are always connected to V_{DD} during this A/D conversion step.

As can be seen in Fig. 7, the positive and negative branches do not operate symmetrically with respect to each other during the error voltage conversion step. This implies that the circuit



(a)



(b)

Fig. 8. Measured distribution of the error voltage $E_{VP.11}$: (a) without LPF; (b) this work.

of Fig. 7 does not work as a differential circuit. It works as a single-ended circuit, which is sensitive to the common-mode noise such as the supply noise or the substrate noise. A temporal averaging scheme is used to reduce the sensitivity of the error voltage $(E_{VP.i})$ measured by using the circuit of Fig. 7(a) to the common-mode noise. The temporal averaging is performed by using a 32-tap FIR LPF.

The operation of the error voltage measurement circuit shown in Fig. 7 is as follows. During the pre-charge phase, the bottom plate of C_5 is connected to V_{DD} and those of $C_4, C_3, \cdots, C_0$ are connected to GND. That is, $(\{D_5, D_4, \cdots, D_0\} = \{‘1', ‘0', \cdots, ‘0'\})$. The evaluation phase is divided into the switching period and the SAR decision period. During the SAR decision (binary search) period of evaluation phase [Fig. 7(b)], D_5 is determined first. If V_{DACN} is smaller than V_{DACP}, D_5 remains at “1.” Otherwise D_5 is switched to “0.” Similarly the other codes $(D_4, D_3, \cdots, D_0)$ are determined such that V_{DACN} approaches V_{DACP}. The final output code of SAR logic is sampled in every 8 clock periods and the sampled value is applied to a LPF for averaging. The LPF averages the 32 consecutive decimated outputs of 5.5-bit SAR logic. The LPF is implemented with a 32-tap comb decimation FIR filter.

To evaluate the sensitivity of the single-ended calibration circuit to the common-mode noise, the error voltage $(E_{VP.11})$ distribution of the MSB capacitor C_{11} of the positive branch DAC is measured, as shown in Fig. 8. The error voltage $(E_{VP.11})$ of Fig. 8(a) is measured by using the 5.5-bit SAR logic of Fig. 7(a). The peak-to-peak variation of the measured error voltage $(E_{VP.11})$ in Fig. 8(a) is 1.5 LSB. This peak-to-peak variation of the error voltage distribution is caused by the common-mode noise such as the supply noise and the substrate noise. To reduce the peak-to-peak variation of the measured error voltage distribution, the LPF (32-tap comb decimation FIR LPF) is used in the error voltage measurement, as shown in Fig. 7(a). As a result, the peak-to-peak variation of the measured error voltage $(E_{VP.11})$ distribution is 0.25 LSB, as shown in Fig. 8(b). The reduction ratio of the peak-to-peak variation of the measured error voltage distribution is 6, which is almost close to the theoretical value of $\sqrt{32}$. The reduced peal-to-peak variation of the measured error voltage distribution shows that the error voltage measurement of Fig. 7(a) is almost robust to the common-mode noise.

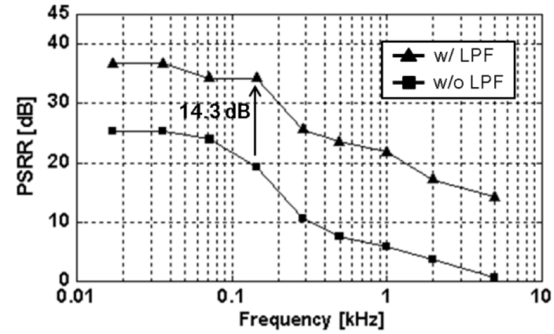


Fig. 9. Measured PSRR of the error voltage measurement circuit of Fig. 7(a).

To show the effectiveness of the temporal averaging scheme to suppress the effect of the common-mode noise on the measured error voltage [Fig. 7(a)], the PSRR of Fig. 7(a) is measured by applying supply noise with and without the temporal averaging scheme. The PSRR is defined to be the ratio of the supply change to the full dc supply voltage divided by the ratio of the output code change to the full scale output code. The measured PSRR is enhanced by 14.3 dB with temporal averaging scheme, which is close to the ideal value of 15 dB $(\sqrt{32})$.

The entire error voltage measurement circuit of Fig. 7 works as a 5.5-bit ADC, whose output ranges from -16 to 15.5 LSB. This ADC range covers the capacitance mismatch ratio $(C_{11} - C'_{11})/(C_{11} + C'_{11})$ of [$-16/2048$, $+15.5/2048$]. Since the extracted code value of $\{D_5, D_4, \cdots, D_0\}$ corresponds to $2E_{VP.11}$ as shown in (11), the code value divided by 2 is stored as $E_{CP.11}$. Thus, the measurable error voltage of the MSB array capacitors ranges from -8 LSB to 7.75 LSB, which corresponds to the capacitance mismatch of [-0.39, 0.38%]. Similarly $E_{CN.11}$ can be extracted from the negative branch split-CDAC. The average value of $E_{CP.11}$ and $E_{CN.11}$ is stored as the final error code $(E_{C.11})$ for the MSB. Since the extracted code value for the C_{10} switching, which is shown in Fig. 6, corresponds to $2E_{VP.10} + E_{VP.11}$ as shown in (13), the $E_{VP.10}$ code is calculated and stored as $E_{CP.10}$. Similarly all the error codes $(E_{C.10}, E_{C.9}, \cdots, E_{C.6})$ are extracted and stored in memory. Since the output of the 5.5-bit ADC is divided by 2, the resolution (step size) of the error code is 0.25 LSB.

In this work, the lower 5.5-bit DAC (LSB array) of one branch is used to convert the residual voltage of the MSB array

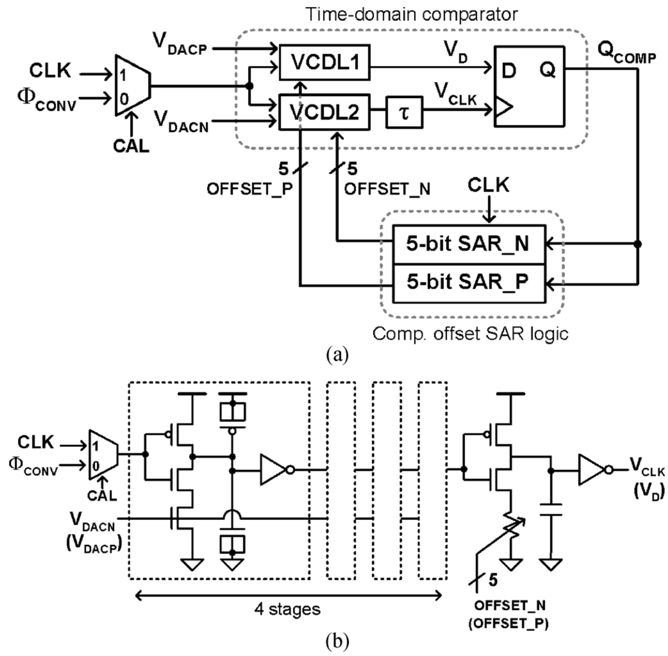


Fig. 10. (a) Time-domain comparator. (b) Schematic of VCDL.

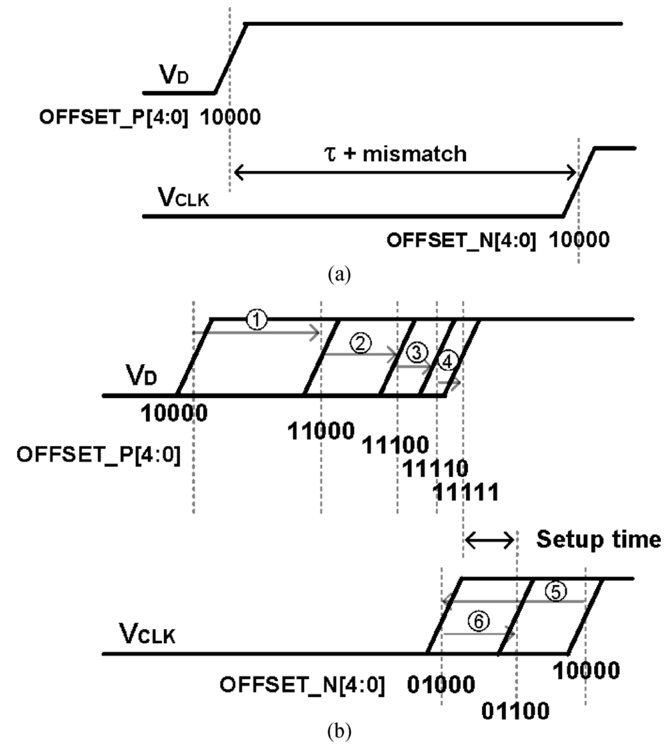


Fig. 11. Comparator offset calibration: (a) after power-on reset; (b) SAR operation for offset calibration prior to error code measurement.

of the other branch. Since the effect of the nonlinearity of the lower 5.5-bit DAC is smaller than 0.5 LSB, it can be neglected in the digital error correction result shown in Fig. 2. In Table I, this effect is 0.22 LSB in the worst case.

C. Time-Domain Comparator With Offset Calibration

A time-domain comparator is used for the comparator in Fig. 4. Compared to the dynamic voltage comparator [17], [18], the time-domain comparator has a low input-referred noise and a small input-offset voltage [9]. Besides, a low supply voltage can be used in the time-domain comparator because it consists of inverters only [6], [9]. In the time-domain comparator shown in Fig. 10(a), the conversion clock Φ_{CONV} is applied $(CAL = 0)$ to two voltage controlled delay lines $(VCDL1, VCDL2)$ in the normal A/D conversion step, and two delayed versions of Φ_{CONV} are applied to the delay (phase) detector DFF (D flip-flop). The delay times of $VCDL1$ and $VCDL2$ are controlled by the analog voltages, V_{DACP} and V_{DACN}, respectively, as shown in Fig. 10(b).

For offset calibration, additional codes $(OFFSET\_P, OFFSET\_N)$ are applied to adjust the VCDL delays, as shown in Fig. 10(b). A dummy delay line τ is inserted in the clock input path of DFF to compensate for the setup time of DFF. The dummy delay line τ consists of an inverter chain, whose delay time is adjusted to be the same as the setup time of DFF (around 1.2 ns at 0.5 V supply).

After the power-on reset, the offset compensation is performed for the time-domain comparator $(CAL = 1)$ by utilizing the comparator-offset SAR logic, as shown in Fig. 10(b). After the offset compensation of comparator is completed, the error codes of DAC capacitors are measured and stored in memory. This ensures that the effect of comparator offset is not included in the stored error code.

For the comparator-offset compensation, the supply voltage V_{DD} is applied to the input nodes (V_{DACP}, V_{DACN}) of comparator. V_{DD} is chosen as the input common mode voltage of the comparator to enhance the accuracy of the measured error codes of DAC capacitors, because the voltage of both input nodes (V_{DACP}, V_{DACN}) of comparator approaches almost V_{DD} at the end of the error code measurement step. At the power-on reset, both control codes $OFFSET\_P$ [4:0] and $OFFSET\_N$ [4:0] are set to "10 000." Then, the comparator-offset SAR logic (5-bit SAR_P) determines the remaining $OFFSET\_P$ [4:0] code from MSB, by one bit in each period of CLK. And then the same SAR operation is performed for the $OFFSET\_N$ [4:0] code for 5 periods of CLK. In this way, the comparator-offset compensation is completed within 10 periods of CLK, and the rising edge of V_D arrives earlier than that of V_{CLK} by around the setup time of DFF, as shown in Fig. 11. The $OFFSET\_P$ [4:0] and $OFFSET\_N$ [4:0] are fixed and do not change after this time. Thus, the equivalent input-offset voltage of comparator is smaller than 0.5 LSB.

IV. Measurement Results

The ADC chip was fabricated in a 0.13-μm standard CMOS process. Fig. 12 shows the layout and micrograph of the fabricated chip. A split-CDAC was implemented by using metal-insulator-metal (MIM) capacitors with the unit capacitance of 100 fF. The minimum allowed value of MIM capacitance is 50 fF due to the DRC rule in the 0.13-μm process used in this work. A margin factor of 2 is taken in this work. Each split-CDAC occupies the chip area of 820 μm $\times$ 230 μm and consists of three sub-capacitor arrays. Two sub-capacitor arrays correspond to the MSB and LSB arrays of Fig. 1. The remaining one sub-capacitor array was implemented for test. The area of split-CDAC is rather large, because the distance between DAC capacitors is widened to reduce the overlapping of routing metal layouts.

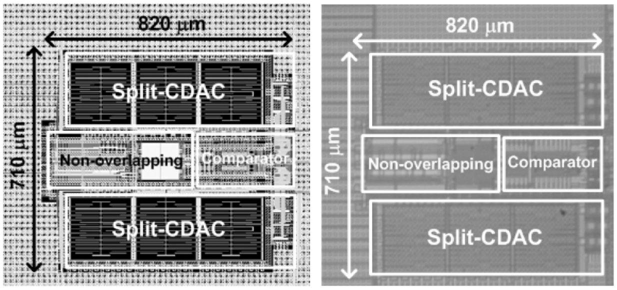


Fig. 12. Layout and micrograph of chip.

The analog block (two split-CDACs, a comparator and a non-overlapping clock generator) consumes 560 nW (post-simulation result) at the supply voltage of 0.5 V, the sampling rate of 10 kS/s and the external clock frequency of 140 kHz. Two split-CDACs, the comparator and the non-overlapping clock generator consume 245 nW, 145 nW, and 170 nW, respectively. The digital block was implemented in a FPGA in this work. The synthesis of the digital block in a 0.13-μm CMOS process showed the power consumption of 170 nW according to *Design Compiler* (*Synopsys*).

The measured error codes $E_{C.i}$ are shown in Fig. 13. Fig. 13(a) shows the contribution of the parasitic capacitance and the mismatch of DAC capacitors to each error code $E_{C.i}$. The parasitic capacitance C_{P2} and C_{P3} are estimated to be 500 fF and 6 fF from the post-layout RC extraction. The contribution of the parasitic capacitance is calculated by using (4). The contribution of the DAC capacitance mismatch comprises higher proportion than that of the parasitic capacitance for all $E_{C.i}$ except $E_{C.10}$.

Fig. 13(b) compares the error codes of positive and negative branches of split-CDAC $(E_{CP.i}, E_{CN.i})$, which are measured in a 0.25 LSB unit. $E_{CP.i}$ is about the same as $E_{CN.i}$ for all i $(i = 11, \cdots, 6)$. Because the positive and negative branch split-CDACs have exactly the same layout, the above observation implies that systematic offset is dominant over the random offset in this work. The maximum error code value of -2.25 LSB at $E_{C.11}$ is located well within the error code range $[-8, +7.75$ LSB] of this work.

Fig. 14 shows the measured DNL and INL for three cases. Without calibration, the missing codes $(\mathrm{DNL} = -1\,\mathrm{LSB})$ occur at eight code locations, as shown in Fig. 14(a). Although the prior work [11] does not show any missing codes without calibration, this work shows missing codes without calibration because a different chip from the same fabrication lot is used in this work. The calibration method of prior work [11] applied to the chip used in this work enhances INL but does not enhance DNL, as shown in Fig. 14(b). The prior work uses the 11-bit raw SAR code for calibration. In this work [Fig. 14(c)], which uses the 11.5-bit raw SAR code for calibration, both INL and DNL are improved with no missing code. DNL and INL are improved by $+0.17/-0.03$ LSB and $+1.47/-1.84$ LSB, respectively, in this work.

The measured frequency spectrum of a 555 Hz sine wave input shows that the calibration of this work suppresses the odd harmonics by more than 12.8 dB. This clearly shows the linearity enhancement by compensating the DAC capacitor mismatch as well as the parasitic capacitance (C_{P2}, C_{P3}).

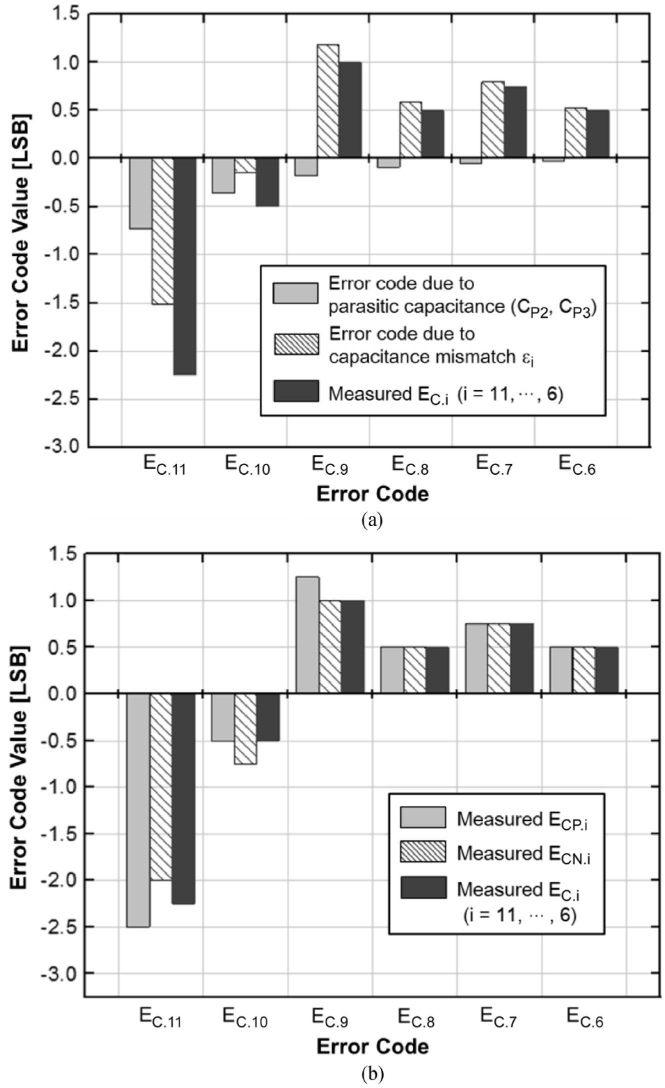


Fig. 13. Measured error code $E_{C.i}$. (a) Effects of parasitic and mismatch. (b) Comparison of positive/negative branch split-CDAC.

Fig. 16 shows the measured SFDR and SNDR with the input frequency up to the Nyquist rate (5 kHz). At the Nyquist rate, SFDR and SNDR are 78 dB and 61.6 dB (ENOB of 9.93 bits), respectively. The figure-of-merit (FoM) is 74.8 fJ/conversion-step, which is obtained by using the measured ENOB and the power consumption including the estimated digital power. Table II compares the performance of SAR ADC with the split-CDAC calibration [5], [7], [8], [12]. The major advantage of this work is that any additional analog circuits are not required for calibration. In the literature, there are not many state-of-the-art SAR ADC papers which have the sampling rate of a few or tens of kS/s. Three papers [19]–[21], which have the CDAC architecture without calibration, are compared with this work in Table III. The FoM of this work is better than those of the conventional CDAC ADC [19], [20], and is worse than that of the split-CDAC ADC [21] without calibration.

V. CONCLUSION

A digital-domain calibration method is proposed to improve the linearity of a differential 11-bit SAR ADC with

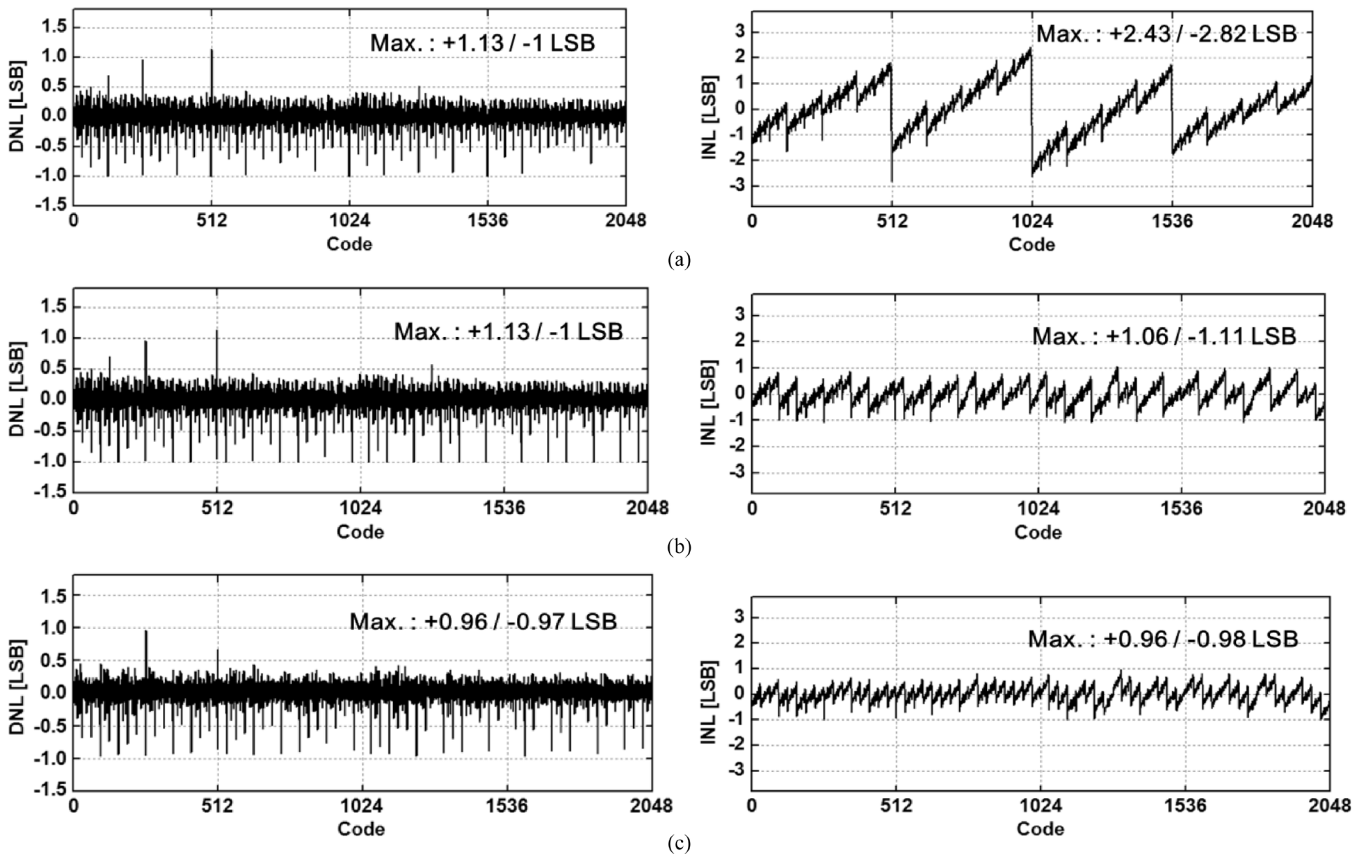


Fig. 14. Measured DNL and INL: (a) without calibration; (b) prior work; [11] (c) this work.

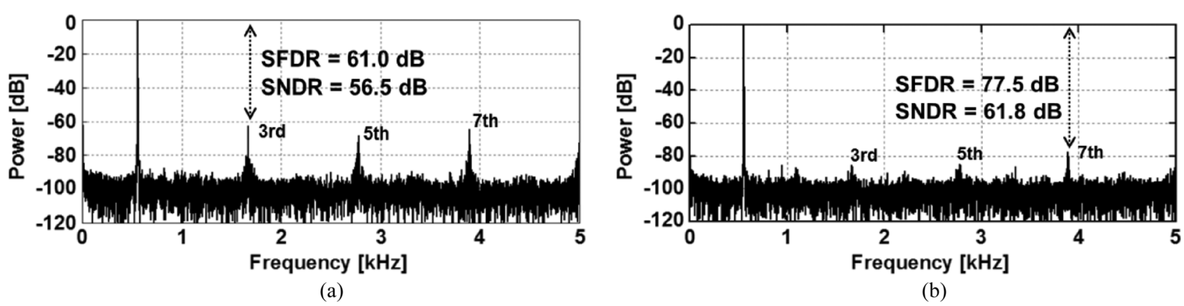


Fig. 15. Measured frequency spectrum of 555-Hz sine wave: (a) before and (b) after calibration.

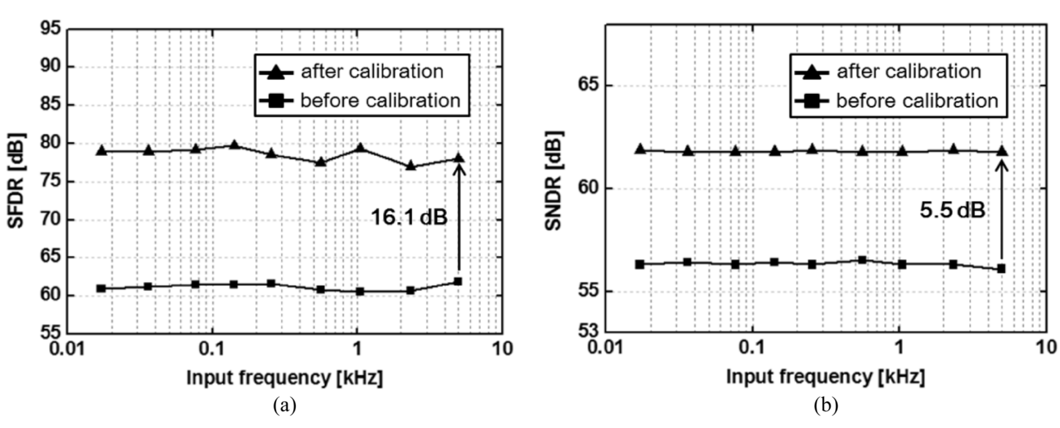


Fig. 16. Measured (a) SFDR; (b) SNDR.

two split-CDACs. It compensates for the nonlinearity caused by the mismatch of DAC capacitors as well as the parasitic capacitances associated with the parasitic capacitors in parallel with the LSB bank and the bridge capacitor, respectively.

TABLE II
COMPARISON OF ADC WITH SPLIT-CDAC CALIBRATION

	ASSCC'07 [5]	CICC'09 [7]	TBCAS'10 [8]	JSSC'12 [12]	**This work**
Technology	0.18 μm	65 nm	65 nm	0.18 μm	**0.13 μm**
Resolution	10 bit	8 bit	10 bit	10 bit	**11 bit**
Supply voltage	1.8 V	1.2 V	1.0 V	Analog : 3.3 V Digital : 1.8 V	**0.5 V**
Sampling rate	1 MS/s	50 MS/s	50 MS/s	768 kS/s	**10 kS/s**
ENOB	8.2 bit	7.3 bit	9.1 bit (@ Nyquist rate)	9.83 bit (@ Nyquist rate)	**9.93 bit (@ Nyquist rate)**
Power consumption	110 μW	1.83 mW	820 μW	58 μW	**730 nW ***
Area	0.052 mm^2	0.03 mm^2	0.039 mm^2	0.01 mm^2	**0.582 mm^2**
FoM	374 fJ/c-s	232 fJ/c-s	29.7 fJ/c-s	74 fJ/c-s	**74.8 fJ /c-s**
Required analog circuit for DAC calibration	Calibration DAC	Tunable capacitor	Tunable capacitor + calibration DAC	None	**None**

(* Analog 560 nW + digital 170 nW)

TABLE III
COMPARISON OF SAR ADC WITH COMPARABLE SAMPLING RATE

	JSSC'03 [19]	JSSC'12 [20]	ESSCIRC'11 [21]	**This work**
Technology	0.18 μm	0.13 μm	0.25 μm	**0.13 μm**
Resolution	8 bit	10 bit	8 bit	**11 bit**
Supply voltage	0.5 V	Analog : 1.0 V Digital : 0.4 V	0.5 V	**0.5 V**
Sampling rate	4.1 kS/s	1 kS/s	31.25 kS/s	**10 kS/s**
ENOB	6.9 bit	9.1 bit (@ Nyquist rate)	7.2 bit (@ Nyquist rate)	**9.93 bit (@ Nyquist rate)**
Power consumption	850 nW	53 nW	87.4 nW	**730 nW**
Area	0.11 mm^2	0.191 mm^2	0.041 mm^2	**0.582 mm^2**
CDAC architecture	Binary-weighted	Binary-weighted	Split	**Split**
Calibration	No	No	No	**Yes**
FoM	1700 fJ/c-s	94.5 fJ/c-s	20 fJ/c-s	**74.8 fJ /c-s**

Any additional analog circuits are not required for calibration, because the error codes for calibration of one split-CDAC are measured by using the other split-CDAC. 6 digital error codes (one for each bit) are measured in sequential steps for each split-CDAC. The digital error codes from two split-CDACs for the same bit are averaged, and the averaged value is stored in memory. During the normal A/D conversion step, the 11.5-bit raw SAR code is added to the fetched error code to generate the calibrated 11-bit code. All the error codes with the corresponding raw SAR code bit of "1" are fetched for the addition. The input offset voltage of comparator is compensated to below 0.5 LSB before the error code measurement step. A time-domain comparator is used as the comparator of this work. The analog blocks (two split-CDACs, a comparator and a non-overlapping clock generator) of SAR ADC were fabricated in a 0.13-μm standard CMOS process. The digital blocks were implemented in a FPGA. The measured SNDR and SFDR are 61.6 dB (ENOB 9.93 bits) and 78 dB at the Nyquist rate with a 5 kHz sine wave input. INL and DNL are measured to be $+0.96/-0.98$ LSB and $+0.96/-0.97$ LSB, respectively. The digital-domain calibration improved DNL, INL, SFDR, and SNDR by $+0.17/-0.03$ LSB, $+1.47/-1.84$ LSB, 16.1 dB, and 5.5 dB, respectively. For the sampling rate of 10 kS/s, the power consumption of the analog blocks is 560 nW at the supply voltage of 0.5 V. The estimated power consumption of digital blocks is 170 nW at 0.5 V according to *Design Compiler*. The estimated FoM of ADC including the assumed power consumption of the digital blocks is 74.8 fJ/conversion-step.

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